Jakarta Indoor $PM_{2.5}$ — All Figures and Tables

March 1, 2026

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1 Main Figures

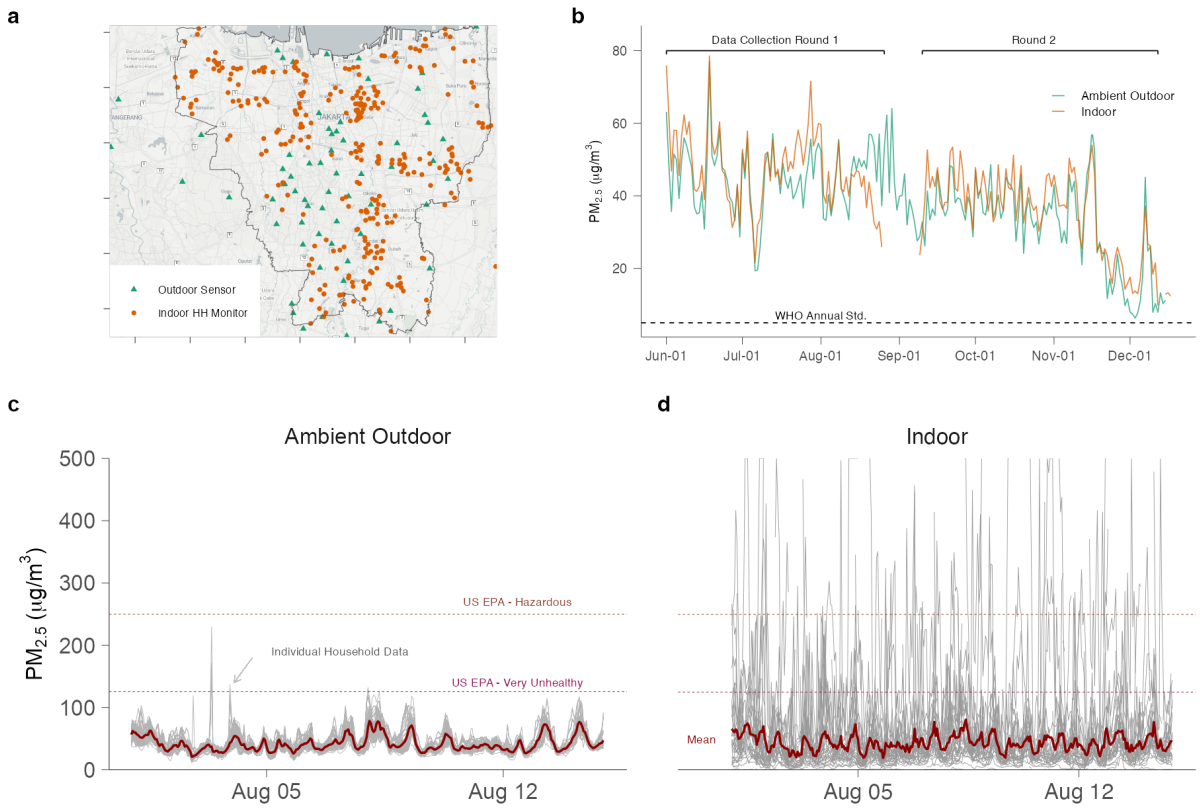

Figure 1: **Figure 1: Indoor and Outdoor $PM_{2.5}$ in Jakarta.** (a) Locations of respondents and installed air quality monitoring hardware. (b) Daily average outdoor and indoor pollution levels. (c) Hourly outdoor $PM_{2.5}$ Aug 1–14, 2024. (d) Hourly indoor $PM_{2.5}$ Aug 1–14, 2024.

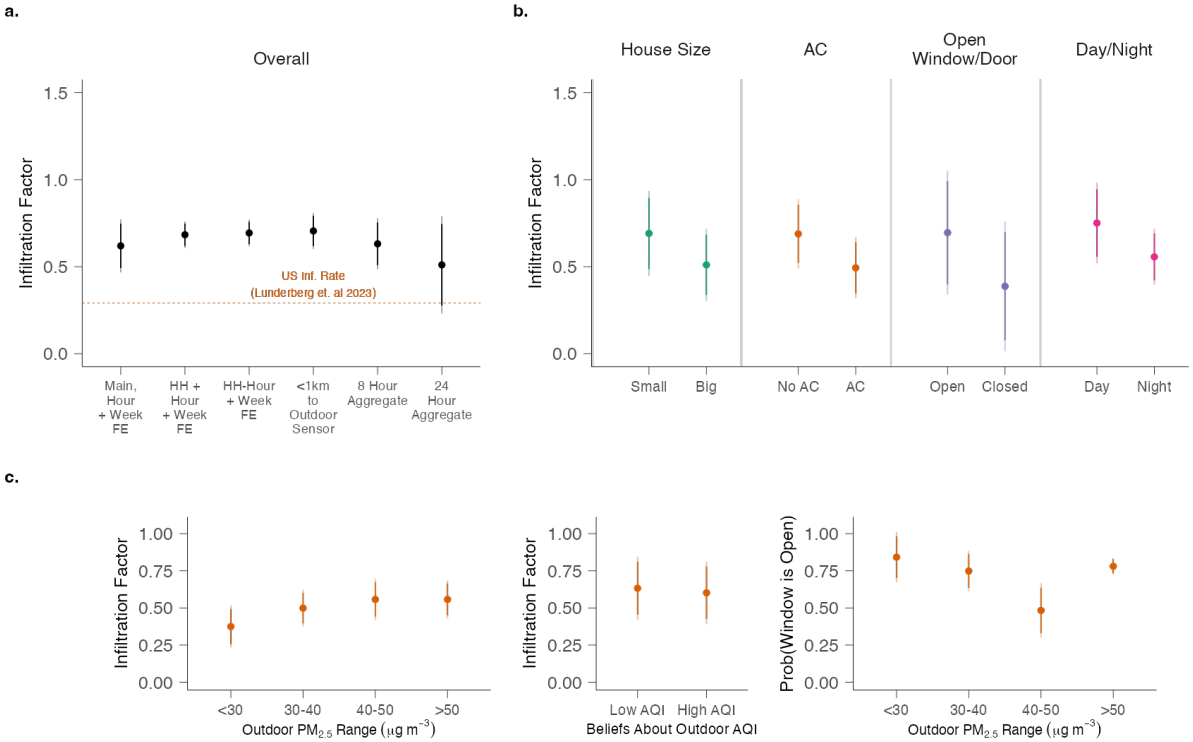

Figure 2: **Figure 2: Infiltration Factor from Ambient Outdoor to Indoor Homes.** (a) Infiltration factors across regression specifications, including the new 24-hour aggregate model. Specifications: (1) Main rhs_fml controls + hour/week FE; (2) HH + Hour + Week FE; (3) HH-Hour + Week FE; (4) <1km to outdoor sensor; (5) 24-hour aggregate, week FE. All regressions upweight hours with higher rates of missing indoor $PM_{2.5}$. (b) Heterogeneity in infiltration factors by household characteristics (income moved to Figure 4). (c) Household behavioral responses to outdoor pollution levels.

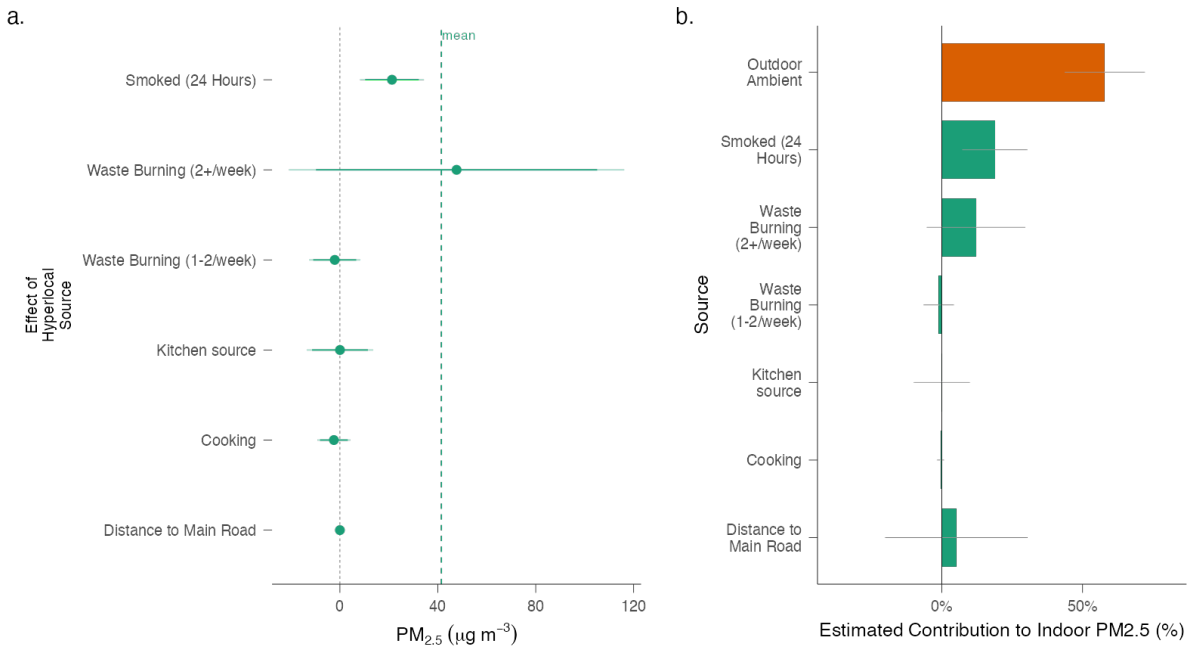

Figure 3: **Figure 3: Contributions of Hyperlocal Pollution.** (a) Regression coefficients of indoor $PM_{2.5}$ on indoor pollution-generating activities: smoking (24 hours), waste burning (1-2/week and >2/week), kitchen/cooking proximity, cooking indicator, and distance to main road. Dashed line shows sample mean indoor $PM_{2.5}$. (b) Mean contribution decomposition: fraction of mean indoor $PM_{2.5}$ attributed to ambient outdoor infiltration versus hyperlocal sources.

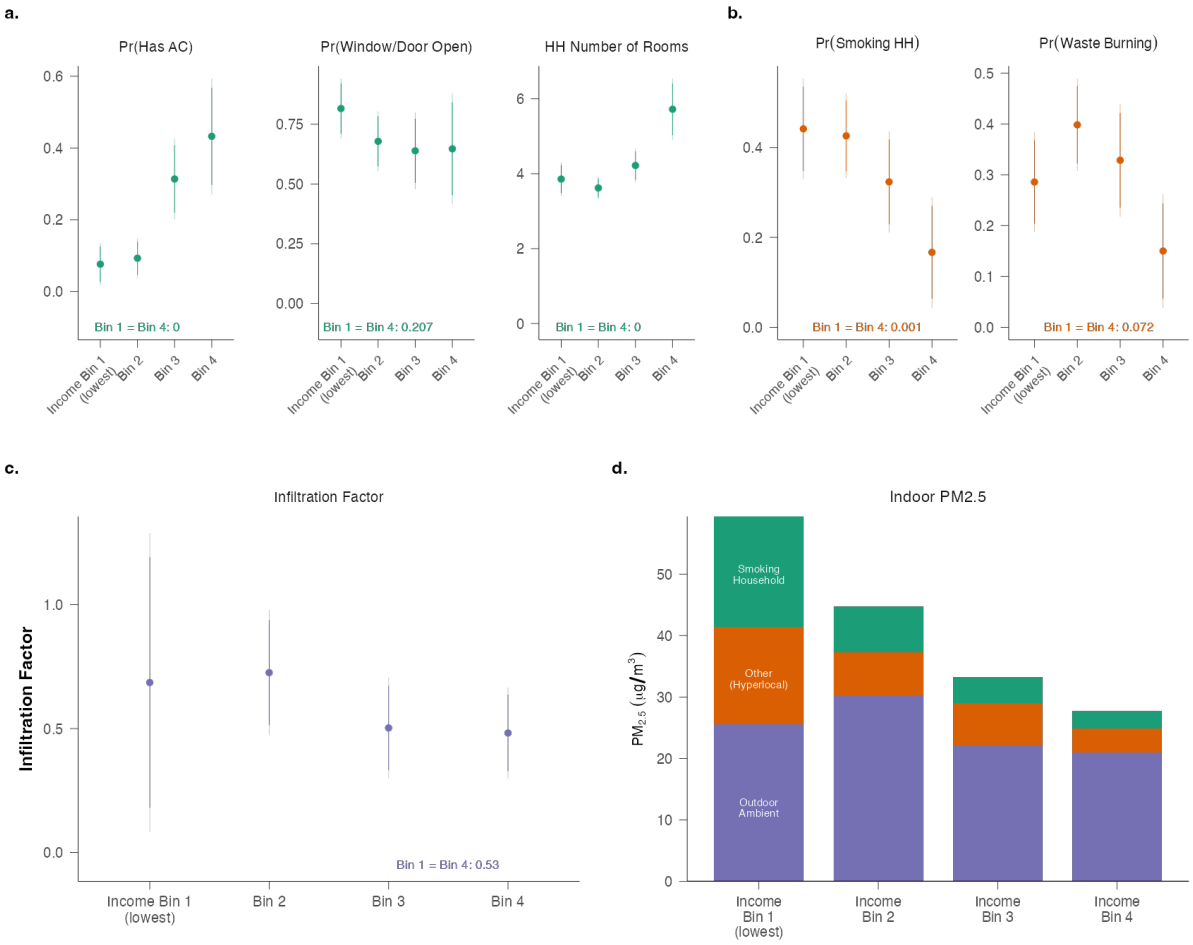

Figure 4: **Figure 4: Indoor $PM_{2.5}$ Higher for Lower Income Households.** (a) Household characteristics by income quartile (AC ownership, window/door open, number of rooms). (b) Hyperlocal pollution sources by income quartile (neighborhood waste burning, indoor smoking). (c) Infiltration factor heterogeneity by income quartile (moved from Figure 2). (d) Decomposition of indoor $PM_{2.5}$ by income quartile into ambient outdoor, smoking, and other (hyperlocal) contributions.

2 Appendix Figures

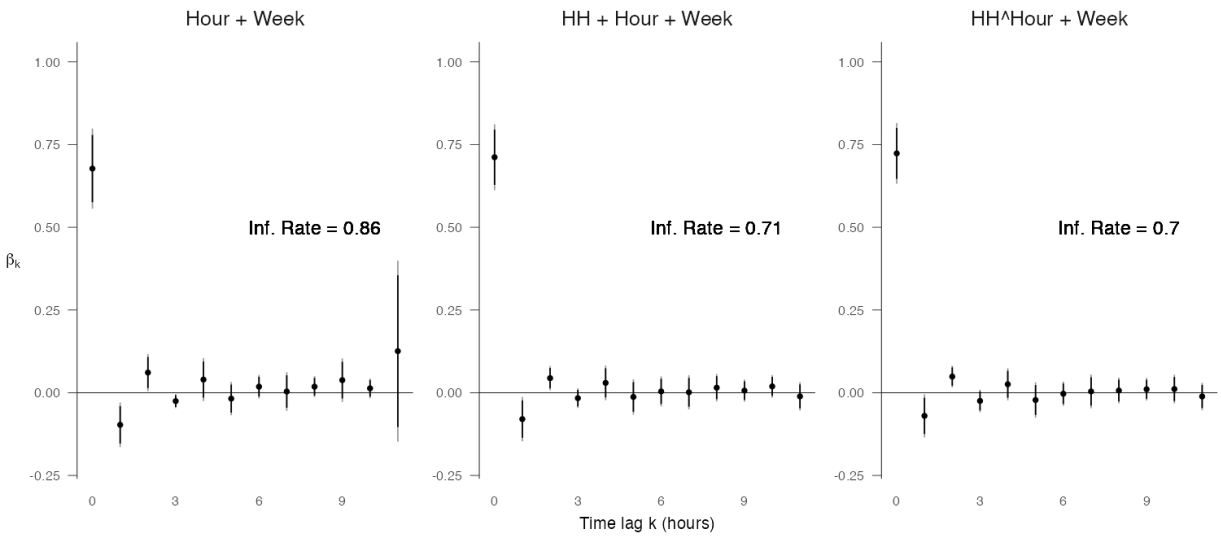

Figure 5: **Figure A1: Infiltration Factor with Lagged Outdoor $PM_{2.5}$.** Distributed-lag specification showing infiltration β_k coefficients for lags $k = 0, \ldots, 11$ hours, across three fixed-effect specifications. The cumulative infiltration factor (sum of β_k) is annotated.

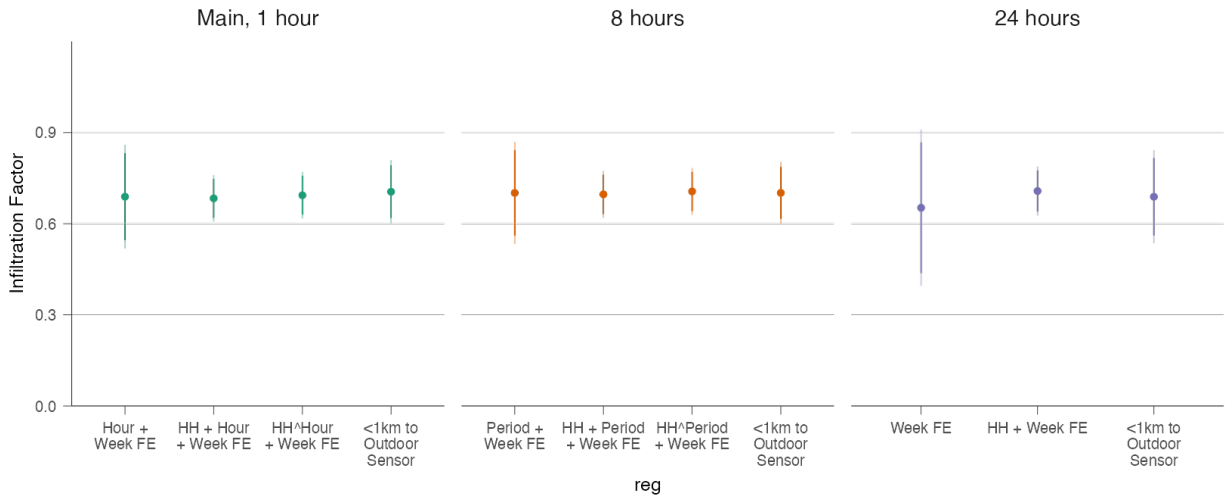

Figure 6: **Figure A2: Infiltration Factor at Different Aggregation Levels.** Infiltration factors estimated at 1-hour, 8-hour, and 24-hour averaging periods, across multiple fixed-effect specifications. Results are consistent across temporal resolutions.

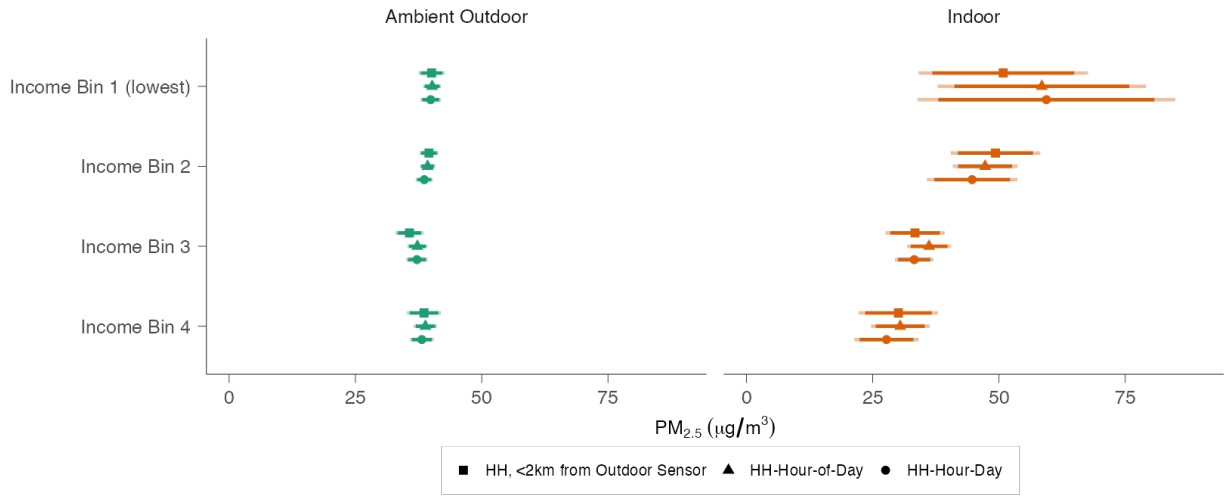

Figure 7: **Figure A3: Robustness — Mean $PM_{2.5}$ Under Various Aggregation Methods.** Mean $PM_{2.5}$ concentrations by income quartile using three aggregation approaches: household-level with <2km sensor proximity, household-hour-of-day averages, and household-hour-date averages.

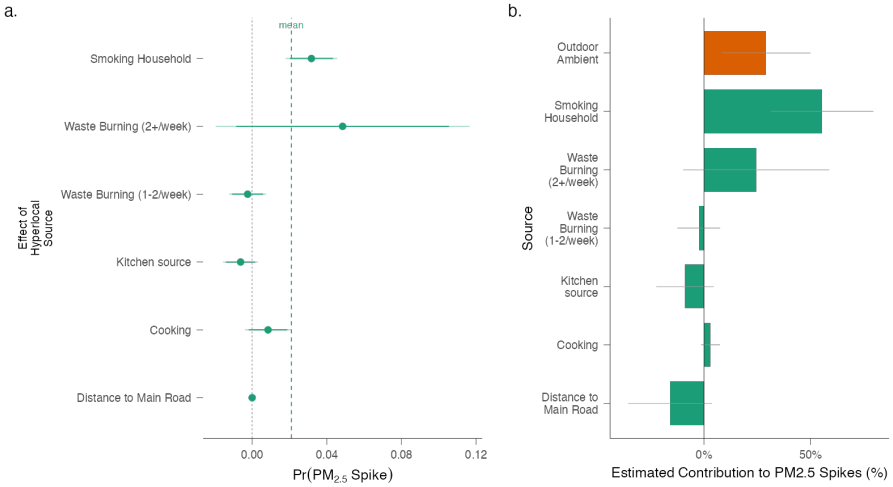

Figure 8: **Figure A4: Source Contributions to $PM_{2.5}$ Spikes.** Mean contribution decomposition with $PM_{2.5}$ spikes (> 125.5 $\mu g/m^3$, the EPA "Very Unhealthy" threshold) as the outcome. Fraction of mean spike probability attributed to each source. Note: this figure is generated after running the updated code.

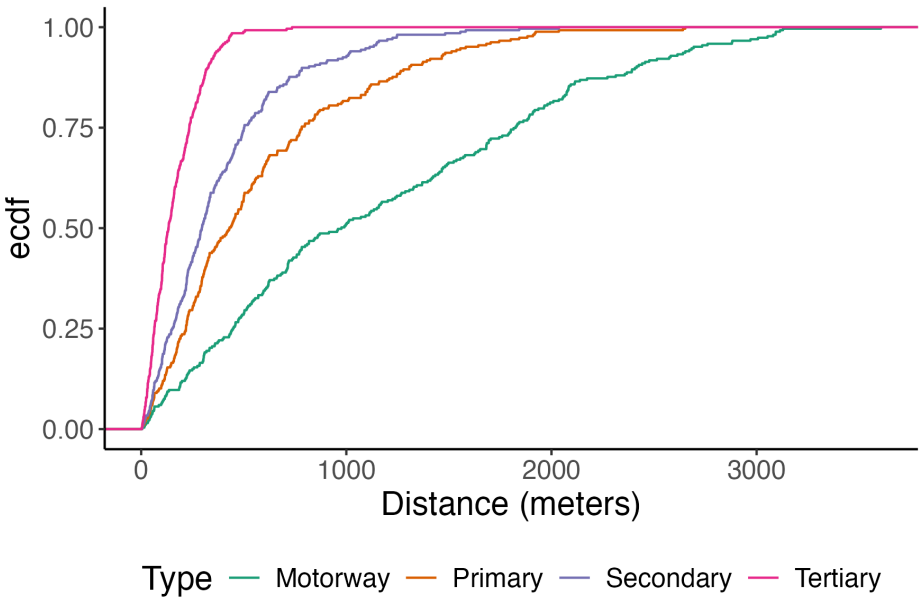

Figure 9: **Figure A5: ECDF of Distance to Roads.** Empirical cumulative distribution of household distance to motorway, primary, secondary, and tertiary roads.

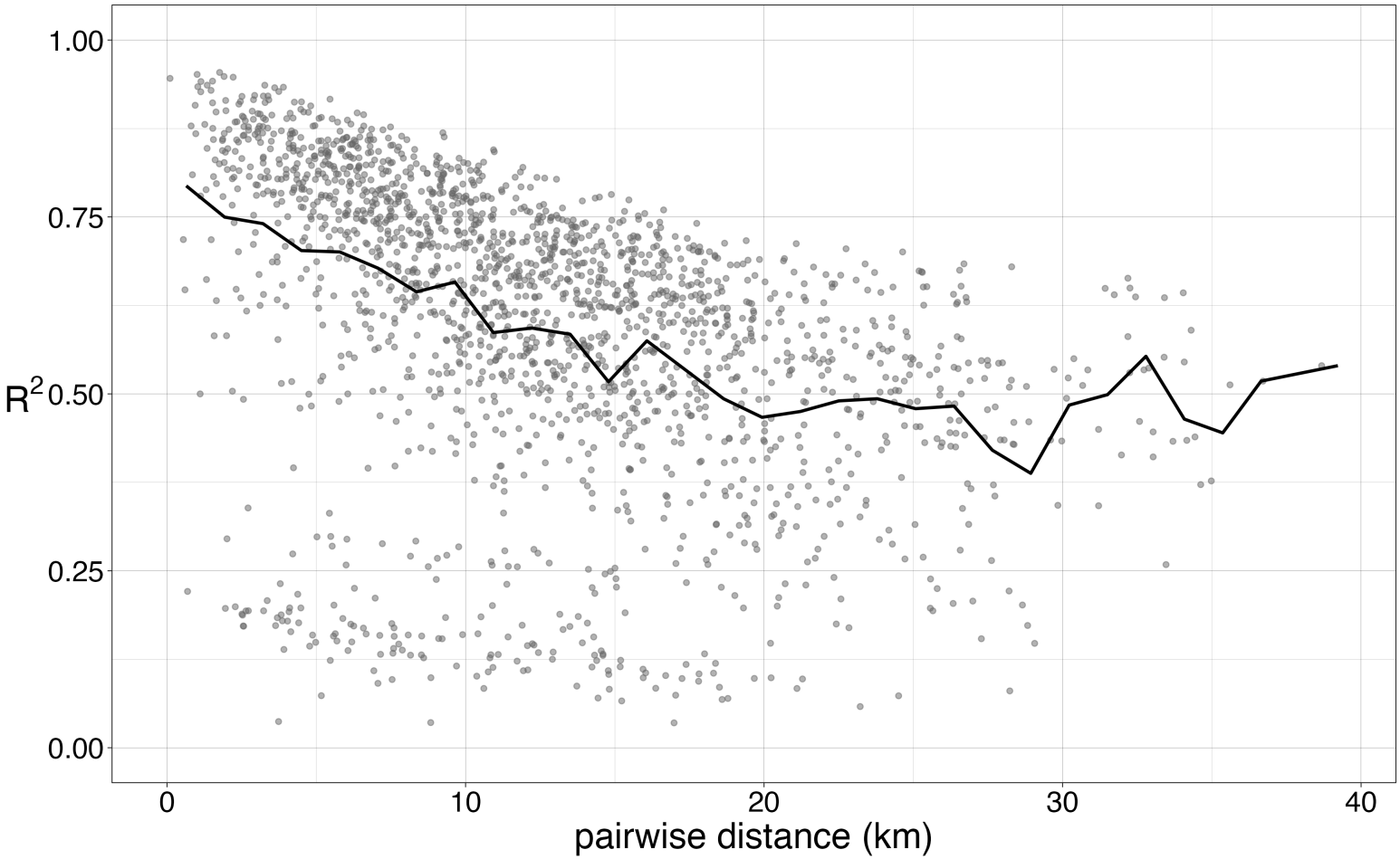

Figure 10: **Figure A6: Spatial Correlation between Outdoor Sensors.** Pairwise R^2 of log-$PM_{2.5}$ between sensor pairs as a function of distance. Outdoor pollution is highly spatially correlated even at distances >10 km.

3 Main Tables

4 Key Results Text

- Hourly concentrations spiked above 100 μg/m^3 during 6.5% of monitored hours.
- Low-income (income_quart = bin 1) households experience 114.2% higher indoor PM2.5 than high-income households (income_quart = bin 4).
- We estimate that households in areas with more than two waste-burning events per week (10.7% of the sample) experienced both higher average indoor PM2.5 concentrations and a higher probability of spikes in indoor PM2.5.
- The probability of measuring spikes in indoor pollution levels was also 341.8% higher in smoking households than in non-smoking households.
- 57.7% of our sample lives within 500 meters from a primary road (dist_primary).
- Overall, we estimate that differences in infiltration explain 14.4% of the difference in average indoor PM2.5 between the lowest and highest income quartile.
- Overall, we estimate that smoking differences explain 47.6% of the difference in average indoor PM2.5 between the lowest and highest income quartile.
- We estimate that differences in other hyperlocal pollution sources explain the remaining 38.0% of the difference in average indoor PM2.5 between the lowest and highest income quartile.

Decomposition of indoor $PM_{2.5}$ sources has been completed and documented using the mean-contribution and income-gap decomposition results shown in Figure 3, Figure 4, and the associated tables.

Source	Reg. Estimate	p.value	Mean Value of Source	Mean Contribution	R2
Outdoor Ambient	0.62	0.000	38.77	58%	
Smoked (24 Hours)	21.28	0.002	0.37	19%	
Waste Burning (1-2/week)	-2.10	0.693	0.21	-1%	
Waste Burning (2+/week)	47.67	0.172	0.11	12%	
Waste Burning					
Kitchen source	0.04	0.996	0.30	0%	
Cooking	-2.45	0.477	0.08	0%	
Distance to Main Road	0.00	0.687	567.08	5%	

5 Appendix Tables

Table 1: **Table A1: Summary Statistics of Household Survey.** Household characteristics from the study sample compared to the Jakarta population mean from SUSENAS 2023.

Variable	Control Mean	N	Population Mean
Female	0.91	308	
Age	41.01	308	
	(6.82)		
HH Size	5.12	308	3.79
	(1.9)		(1.31)
Number of Children	2.05	307	
	(1.01)		
HH Monthly Exp. (USD)	248.28	308	655.94
	(115.96)		(707.71)
HH Monthly Income (USD)	308.58	308	
	(191.39)		
HH Head Attended Secondary	0.88	308	0.88
HH Head Attended Tertiary	0.13	308	0.3
Owns AC	0.54	292	0.39
	(0.5)		
LPG Stove	0.98	307	0.96
Number of Smokers	0.93	308	0.73
	(0.84)		(0.71)

Table 2: **Table A2: Distance to Outdoor Sensor Not Correlated with Household Characteristics.** Regression of household distance to nearest outdoor sensor on household covariates. No evidence of selection bias on observables.

Dependent Variable:	Distance to Outdoor Sensor (meters)
Model:	(1)
Variables	
Indoor PM2.5	-1.45
	(2.64)
Outdoor PM2.5	3.43
	(9.79)
Income (USD)	0.17
	(0.65)
Housing Size	41.92
	(59.20)
AC	-126.77
	(216.44)
HOH Employed	424.34
	(411.19)
HOH Attended Secondary School	-392.95
	(339.96)
HOH Attended Tertiary School	-305.85
	(348.82)
Fit statistics	
Dependent variable mean	2,380.7
Observations	267

IID standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Table 3: **Table A3: Waste Burning Not Correlated with Higher Ambient Outdoor $PM_{2.5}$.** Regression of outdoor $PM_{2.5}$ on reported neighborhood waste burning frequency, for all households and restricting to households within 2km of a sensor.

Dependent Variable:		Outdoor PM2.5
	Closest 3 Sensors	HH Within 2km of Outdoor Sensor
Model:	(1)	(2)
Variables		
Waste Burning (1-2/week)	0.586	0.272
	(0.755)	(0.757)
Waste Burning (2+/week)	3.83***	5.49***
	(1.04)	(1.94)
Outdoor Temperature	0.329	0.745***
	(0.266)	(0.271)
Outdoor Humidity	-0.206***	-0.198***
	(0.042)	(0.050)
Fixed-effects		
Week FE	Yes	Yes
Hour FE	Yes	Yes
Fit statistics		
Observations	449,568	230,234
R^2	0.35005	0.33996
Within R^2	0.01325	0.02104

Clustered (respondent_id & date_hour) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Table 4: **Table A4: Source Contributions to $PM_{2.5}$ Spikes.** Analogous to Table 1 but with $PM_{2.5}$ spikes ($> 125.5\ \mu g/m^3$) as outcome.

Source	Reg. Estimate	p.value	Mean Value of Source	Mean Contribution	R2 Contribution
Outdoor Ambient	0.000	0.006	38.77	29%	2%
Smoking Household	0.032	0.000	0.37	55%	53%
Waste Burning (1-2/week)	-0.002	0.628	0.21	-2%	
Waste Burning (2+/week)	0.048	0.161	0.11	24%	
Waste Burning					21%
Kitchen source	-0.006	0.194	0.30	-9%	2%
Cooking	0.009	0.177	0.08	3%	1%
Distance to Main Road	0.000	0.112	567.08	-16%	5%